

Appendix C: SCOPE OF WORK

1 Objectives

The objective of this engagement is to establish a modern, data-driven assurance Solution leveraging advanced analytics and AI capabilities to strengthen oversight of contractor costs and payment processes. The Solution is intended to:

- Ensures transparency of contractor reported costs under a target price payment regime;
- Provides reliable verification of incurred costs;
- Enables early detection of anomalies, irregularities, or emerging cost overruns;
- Delivers independent, evidence-based assurance to the Corporation and its governance bodies;
- Minimizes manual effort through automation, advanced analytics and AI-driven processing; and
- Supports timely, defensible, and well-documented payment decisions.

Through this NRFP, the Corporation seeks a Proponent with demonstrated expertise in AI-enabled commercial assurance solutions, associated technologies, and supporting services on an “as-and-when” needed basis, required to successfully design, implement, configure, and support the Solution. The intent is to establish a comprehensive, scalable, and future-ready approach to commercial oversight that combines technology, analytics, and subject matter expertise to support effective project delivery.

2 Current State

The Corporation administers complex, multi-year capital projects involving numerous vendors, evolving contract terms, and high volume of invoices. Cost verification is critical to ensuring contract compliance, financial control, and public accountability.

Currently, invoice and cost verification processes are standardly manual and require reviewers to cross-reference invoices against executed contracts, amendments, rate cards, and schedules using multiple document formats. Vendor invoices and relevant supporting documents are inconsistent in structure and detail, increasing review time and the risk of error.

As the relevant projects scale, the current approach presents challenges such as:

- Manual effort and limited scalability, resulting in extended review cycles;
- Inconsistent review outcomes due to reliance on individual judgment rather than standardized, automated controls;

- Increased risk of value leakage, including overbilling, duplicate charges, and non-compliant costs; and
- Limited visibility into invoice compliance and emerging cost issues.

Existing systems support transactional processing but do not provide automated, end-to-end validation of the payment processing documentation against current contract terms or consolidated reporting for audit-ready traceability.

As invoice volumes and contract complexity increase, the Corporation's current cost verification model faces efficiency, risk management, and oversight limitations. There is a clear need to modernize this function through a scalable, AI-enabled Solution that enhance accuracy, consistency, and proactive cost control.

3 Scope of Services

The Proponent's Proposal is expected to describe a complete and integrated Solution that satisfies all requirements set out in this NRFP. The Corporation has established the following categories of requirements:

- Business Requirements (Section 5),
- Technical Requirements (Section 6), and
- AI Governance Requirements (Section 7).

Collectively, these requirements establish the minimum acceptable standard for the proposed Solution. The Corporation expects the Proponent to clearly demonstrate, in measurable and substantiated terms, how its proposed Solution meets or exceeds the requirements within each category.

The Proponent is expected to ensure that the:

- proposed AI-enabled assurance tool,
- Proponent's qualifications and experience, and
- Proponent's overall delivery approach

collectively provide a comprehensive, secure, scalable, and effective Solution that satisfies all applicable requirements of this NRFP. The Corporation expects the Proponent to demonstrate how the proposed Solution, supporting services, governance framework, and delivery approach will work together to support a practical, reliable, and future-ready commercial assurance capability.

In addition to supplying an AI-enabled tool for cost verification, the Proponent is expected to demonstrate the capability to deliver additional supporting services on an “as-and-when” needed basis that complement and strengthen the Solution. The Corporation requires a Proponent with proven experience in the provision of services relating to the delivery of major capital projects, and not solely in payment processing or invoice review. This includes demonstrated expertise in:

- commercial management which may include dispute resolution and fraud investigation services,
- contract interpretation,
- cost and schedule analysis, and
- the broader business processes that underpin large, complex infrastructure projects within the context of this NRFP.

4 Delivery Model

Proponents are invited to propose their Solution under the following delivery model:

Managed Service Model:

The successful Proponent delivers cost verification outcomes as a fully managed service, including data ingestion, analysis, reporting, and quality assurance. TI Corp consumes the outputs and owns the data, while the successful Proponent retains ownership and operation of the underlying platform and tooling.

5 Business Requirements

The Proponent is expected to provide an AI-enabled assurance Solution that automates the majority of cost verification, invoice validation, anomaly detection, and advanced analytics for infrastructure projects delivered under a Target Cost contracting model. The following business requirements apply.

5.1 Automated, Continuous Cost Verification and Analysis

The Solution from the Proponent is expected to automatically verify contractor reported costs for labor, plant and equipment, subcontractors, materials, indirect costs, overhead, and fee calculations. All costs should be validated against the contractually defined payment conditions and terms (such as allowable and disallowed costs, and fixed fee rates). In addition, the Solution is expected to:

- Flag exceptions, anomalies, or unsupported charges for human review; and
- Provide structured outputs suitable for payment processing and audit trails.

The Solution is expected to perform these primary verification functions, with human review required only for exceptions or complex cases.

5.2 Invoice and Ledger Automation

The Solution is expected to ingest data from the relevant sources and deploy AI-enabled capabilities in order to:

- Crosscheck invoices, timesheets, equipment logs, and any other supporting documentation;
- Identify duplicate charges, misclassification of cost items, and general inconsistencies;
- Reconcile invoiced amounts with approved work packages, cost breakdown structures, and earned progress; and
- Generate automated audit logs and exception reports.

The Solution is expected to provide structured outputs that support timely payment decisions and reduce manual audit effort.

5.3 Change Order Management and Validation

Change orders often involve complex pricing, rate application, and substantiation requirements, and inconsistent manual review can lead to value leakage, delays, or disputes. The Solution is expected to streamline and standardize the end-to-end change management process, ensuring that all changes are assessed in accordance with contractual requirements and supported by appropriate documentation. The Solution should be capable of:

- Automating the development and review of change orders, including but not limited to review of rate application and pricing, cost substantiation, request for written approval prior to execution, and issuance of change certificate.
- Flagging inconsistencies, unsupported costs, or deviations from contract requirements.
- Providing structured outputs to support commercial decision-making.

The AI-enabled tool should perform the primary analytical and validation functions, with Proponent's human review activities only focused on exceptions or complex commercial judgements. The Proponent's expertise, when required, should complement the tool by interpreting nuanced contractual provisions, advising on commercial implications, and ensuring that change requests are managed in a manner consistent with best practices for large, complex infrastructure projects.

Approval of change orders and interpretation of contractual entitlement remain management responsibilities of the Corporation.

5.4 Monitoring of Target Cost vs. Contract Price

The Solution should support continuous monitoring of key commercial metrics, including:

- Actual cost versus contract price (which consists of target cost and other relevant assumptions based on the contractual payment mechanism);
- Exposure to gainshare/pain share and other incentive mechanisms when applicable; and
- Estimate-to-complete and estimate-at-completion forecasts and trends.

Monitoring should be automated wherever possible, with dashboards or reports generated by the Solution.

5.5 Predictive Analytics (Optional)

This functionality is optional and will **not** be evaluated as a value-added feature. However, the Corporation may elect to implement or expand this functionality in the future, subject to project requirements and written direction or approval.

The Proponent is expected to provide AI-driven predictive analytics capable of generating:

- Probabilistic cost forecasts;
- Risk and drawn contingency values considering assumptions provided by the project team; and
- Early warning alerts for emerging cost overruns, productivity issues, or anomalies.

Models may incorporate inputs such as schedule progress, earned value metrics, productivity data, commodity price trends, and other relevant indicators. The objective is to provide forward-looking assurance, not only retrospective verification.

5.6 Procurement and Subcontract Review (Optional)

This functionality is optional and will **not** be evaluated as a value-added feature. However, the Corporation may elect to implement or expand this functionality in the future, subject to project requirements and written direction or approval.

The Solution's functionality would be expected to provide AI-driven assessment relating to:

- Analysis of subcontract pricing patterns and variances, and
- Benchmarking against market pricing.

This optional functionality is intended to enhance commercial transparency and market intelligence.

6 Technical Requirements

The Proponent is expected to provide a technically robust Solution and service approach that supports the overall requirements of this in Section 3 and 4.

The Solution should align with the Corporation's requirement for strong oversight, auditability, transparency, and data sovereignty, and must operate effectively within a constrained internal IT and AI capability environment.

Mandatory Technical Requirements

Proponents are to refer to Appendix D – Proponent Response Worksheet, tab “Mandatory Requirements” for the technical criteria that must be met.

Scored Technical Requirements

Proponents should refer to Appendix D – Proponent Response Worksheet, tab “Technical Requirements” for the detailed scored technical requirements.

In addition to the mandatory and scored technical requirements, Proponents are requested to provide a written response, not exceeding one (1) page per topic, addressing the key considerations identified below. These responses are intended to provide additional context regarding the Proponent's approach, methodology, experience, and overall Solution capabilities.

6.1 Integration and Data Connectivity

- Proponents should describe how their Solution supports integration with Project Contractor systems and the Corporation systems required for cost verification, including ERP, accounting, cost management, project controls, and document repositories.
- Proponents should clearly describe how data is ingested, processed, and synchronized across systems, including assumptions related to data availability, structure, and quality.
- Integration should be implemented using standard, secure methods such as APIs, secure file transfer, or batch processing, and must be designed to operate consistently across multiple projects.

6.2 Processing Model and Operational Flow

The Proponent should describe the end-to-end processing model for cost verification activities, including how data flows through the system or service from ingestion to validation, exception handling, and reporting.

This description should clearly identify:

- which activities are automated;
- which activities are rule-based;
- which are supported by analytical or AI techniques; and
- which require human review.

The Proponent should also describe how urgent or high-risk issues are handled, including the ability to:

- escalate findings;
- trigger immediate investigation workflows; and
- engage qualified specialists where required.

6.3 Reporting, Auditability, and Traceability

The Solution should generate outputs that are fully traceable, auditable, and suitable for governance, oversight, and dispute resolution. This includes the ability to link outputs to underlying data, applied rules, and review actions, and to reproduce results where required.

Reporting should support both:

- operational use (reviewers, analysts); and
- governance use (executive, audit, oversight bodies)

6.4 Data Ownership Requirements

The Proponent must comply with all applicable Connected Services British Columbia (CSBC) policies and requirements, and the Corporation requirements including but not limited to:

6.4.1 Data Residency (Mandatory Requirement)

This is part of the Mandatory Requirements set out in the NRFP document.

- Data must be stored and processed within Canada unless otherwise approved in writing by the Corporation.

- Data must not be accessed, stored, or transmitted outside Canada without prior written approval.

6.4.2 Data Ownership and Access (Mandatory Requirement)

- Data, metadata, logs, analytics outputs, reports, and derived insights must remain the exclusive property of the Corporation.
- The Corporation must retain full access to and control over data at all times.
- The Solution must allow the Corporation to export data in a structured, machine-readable format without additional cost.
- The Proponent must not use the Corporation data to train or improve internal, external, or shared models without the Corporation's explicit written consent.

6.4.3 Data Portability (Mandatory Requirement)

The Solution **must** ensure full data portability to support long-term flexibility and avoid dependency on a single vendor. The Solution must allow the Corporation to export data, analytical results, and system configurations in a complete and usable format including but not limited to, PDF, word, excel and CSV. All exports must be provided in commonly accepted, non-proprietary formats that enable reuse and migration. Solutions that rely on proprietary data structures or formats that limit the Corporation's ability to access or transfer its data will not be considered acceptable.

6.4.4 Data Preparation Responsibilities

Proponents are expected to clearly define their approach to data preparation and management as part of their Solution. This includes identifying all required data inputs, expected data formats, and minimum data quality standards. Proponents should also clearly state responsibilities for data extraction, transformation, and validation, including any dependencies on the Corporation or third parties. Where responsibilities are shared, this should be explicitly defined to ensure clarity and accountability throughout implementation and operation.

6.4.5 Architecture Alignment

The Solution should align with the Corporation's enterprise architecture principles and technology standards. The Solution is expected to:

- Support deployment within a Microsoft Azure-aligned cloud environment
- Integrate with the Corporation's broader data and analytics ecosystem
- Support reporting and visualization tools such as Power BI; and

- Support scalability, modularity, and future integration with enterprise system.

6.5 Technical Support and Maintenance

The Proponent is expected to:

- Provide a technical support and maintenance model.
- Provide operational support for system availability, data processing, validation execution, exception handling, and reporting.
- Maintain monitoring, logging, and incident management processes to support reliability, traceability, and timely issue resolution.
- Define service levels for operational activities, including processing timelines, system availability, and incident response.
- Support continuous improvement of validation rules, analytical models, and evolving contract and data requirements.
- Ensure data, rule sets, configurations, and audit records remain accessible and portable.
- Provide ongoing maintenance and technical support, including system updates, security patching, performance optimization, and issue resolution.

7 AI Governance Requirements

7.1 AI Capability Requirements

Proponents should clearly describe their AI capabilities, including both strengths and limitations, as part of their submission. At a minimum, the proposed Solution should support the automated or semi-automated classification and validation of cost data, including the ability to process large volumes of transactions. The Solution should provide cost assessment capabilities based on defined rules and/or pattern recognition techniques.

7.2 AI Transparency and Control

All AI-enabled outputs should be traceable to their underlying data sources. The Solution should clearly identify the logic, rules, or models used to generate outputs and should indicate the data inputs on which those outputs are based.

The Solution should support evidence-based assurance processes by ensuring that all outputs can be reviewed, validated, and reproduced where required. Proponents should demonstrate how their Solution maintains transparency and avoids reliance on opaque or non-traceable processes. For each flagged or rejected cost, the Solution should produce a clear, human-readable reason referencing contract clauses, rules, or detected patterns.

7.3 AI Operating Model

Proponents should clearly describe how their Solution operates across different modes of processing. This includes distinguishing between automated processing, rule-based validation, and manual review activities.

Proponents should also disclose the expected level of automation within their Solution, including an estimate of the percentage of transactions or processes that can be handled without manual intervention. In addition, Proponents should identify areas where human involvement is required, including complex scenarios, exception handling, and final validation. Additionally, the proponents' AI should be aligned with BC Gov's generative AI use policy:

<https://digital.gov.bc.ca/policies-standards/generative-ai-policy/> and responsible use principles: <https://digital.gov.bc.ca/ai/draft-responsible-use-principles/>).

References:

The Canadian Center for Cybersecurity issued recommended controls for AI security: <https://www.cyber.gc.ca/en/guidance/top-10-artificial-intelligence-security-actions-primer-itsap10049>.

8 Privacy and Security Requirements

The Proponent must ensure the Solution aligns with applicable privacy, security and data governance requirements.

Proponents are required to refer to Appendix D – Proponent Response Worksheet, tabs “Mandatory Requirements” and “Privacy & Security Requirements” and provide detailed responses to the specified privacy and security criteria.

9 Deliverables

The Solution should deliver, at a minimum, the following outputs:

- An implemented AI-enabled solution for automated cost verification, validation, and analysis;
- Configured data integrations, validation rules, workflows, and analytical capabilities
- Verified cost outputs, exception reports, and audit-ready records;
- Dashboards and reporting to support payment processing and commercial oversight
- Exportable reports in standard formats (refer to Section 8.3);
- Implementation and transition; and
- Ongoing operational support, maintenance, and continuous improvement.

Detailed deliverables, scope, and responsibilities will be finalized during the negotiating and contracting phase.

10 Implementation and Timeline

The Proponent should provide an implementation approach and timeline for deployment of the Solution, including key phases, milestones, deliverables, dependencies, and resource requirements. The implementation approach should include, at minimum:

- Project initiation and requirements confirmation;
- Data assessment, integration, and configuration;
- Development and testing of workflows, validation rules, and analytical models;
- User acceptance testing and training;
- Production deployment and go-live support; and
- Post-implementation support, optimization, and knowledge transfer.

Proponents should also identify:

- estimated implementation timelines;
- key assumptions and risks;
- required Corporation involvement; and
- any proposed phased deployment or transition approach.

11 Pricing

Proponents must provide separate and clearly identifiable pricing in the pricing response worksheet (Appendix E).

Pricing submissions should clearly identify all proposed costs, including breakdown and assumptions, which may include implementation, configuration, licensing or subscription fees, ongoing support, and additional supporting services when needed.